

Use Case Specification: Book Event Ticket

1. Overview

- **Use Case ID:** UC-01
- **Use Case Name:** Book Event Ticket
- **Primary Actor:** Attendee

2. Preconditions and Postconditions

Preconditions:

- The attendee is logged into their account.
- The event is published, active, and has available tickets.

Postconditions:

- The ticket inventory is decremented by the purchased quantity.
- A digital ticket with a unique ID and QR code is issued to the attendee.
- A confirmation email with the ticket receipt is sent.

3. Main Success Flow

1. **Search Events:** Attendee searches or filters active events.
2. **View Listings:** System displays matching active events.
3. **Select Event:** Attendee selects an event from the list.
4. **Display Details:** System shows event details, pricing, and seat availability.
5. **Choose Quantity:** Attendee selects ticket quantity and clicks Checkout.
6. **Reserve Seats:** System verifies capacity and temporarily holds the tickets.
7. **Enter Details:** Attendee enters or confirms contact information.
8. **Review Order:** Attendee reviews the order summary and total cost.
9. **Submit Payment:** Attendee enters payment details and submits the order.
10. **Authorize Payment:** Payment gateway processes and approves the charge.
11. **Update Inventory:** System permanently deducts the tickets from event capacity.
12. **Generate Ticket:** System creates a digital ticket with a unique ID and QR code.
13. **Send Receipt:** System emails the digital ticket and purchase receipt.
14. **Display Confirmation:** System displays the order confirmation screen.

4. Alternate Flow: Payment Authorization Fails

Branches from Step 10 of the Main Flow when payment is declined.

- 10a.** Payment gateway declines the transaction.
- 10b.** System releases the held tickets back to available inventory.
- 10c.** System displays a payment error message.
- 10d.** Attendee is prompted to retry with different details or cancel.